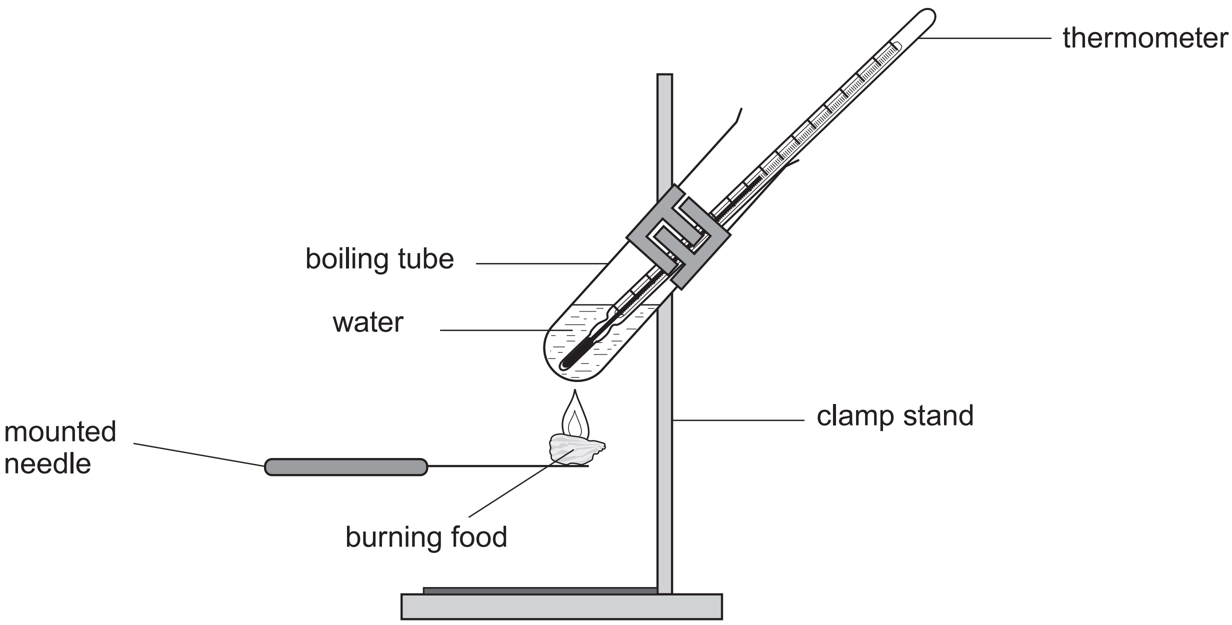
Examiner only

4. The energy your body needs comes from the food you eat. The energy content of food can be measured in the laboratory.

(a) The apparatus below can be used to measure the energy content of food in a school laboratory.



(i) State **two** variables which should be controlled in this experiment **and** explain why they need to be controlled. [2]

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(ii) Explain why the mean results obtained in the school laboratory are different from manufacturers' values given on a food label. [2]

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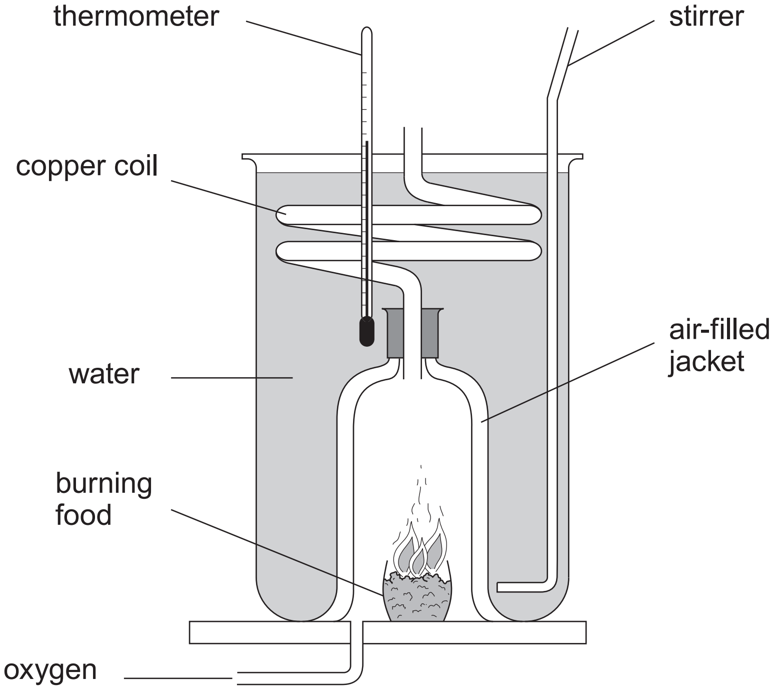
Examiner only

(iii) The energy content of food can also be measured using the calorimeter shown in the diagram below.



Explain why using this calorimeter is likely to improve the quality of the results. [3]

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(b) The energy absorbed by a water sample is given by the equation

energy (J) = mass of water (kg) × temperature rise (°C) × 4200

Results from the experiment for one food sample are shown below.

- mass of water = 20.0 g
- temperature rise = 15.4 °C
- mass of food sample = 12.4 g

Use the information above to calculate the energy content of the food in J/g. [3]

1g = 10^{-3} kg

energy content = ... J/g

10

